22684251-trinhduonghoan

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[2]:
```
import matplotlib.pyplot as plt
import numpy as np
import cv2
```

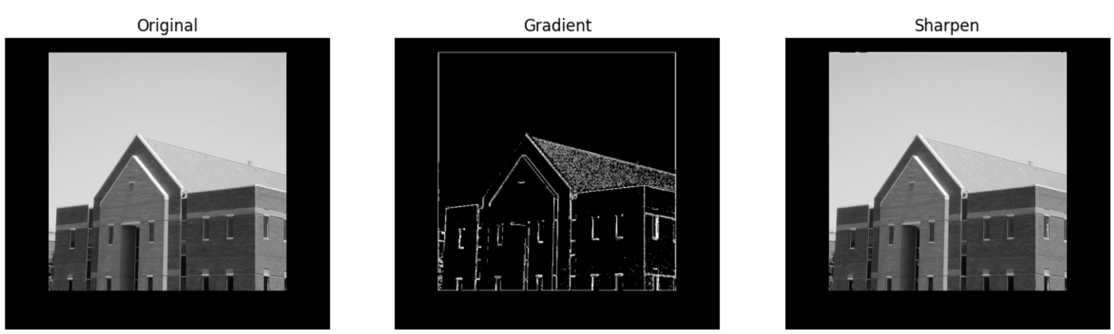
[3]:
```
list_path_image = ['building.jpg', 'cameraman.png', 'high-exposure.jpg', 'lena.
↪jpg', 'low-exposure.jpg']
```

[4]:
```
kernel_x = {
    'Robert': np.array([
        [0, 0, 0],
        [0, 1, 0],
        [0, 0, -1]
    ], dtype=np.float64),
    'Prewitt': np.array([
        [-1, 0, 1],
        [-1, 0, 1],
        [-1, 0, 1]
    ], dtype=np.float64),
    'Sobel': np.array([
        [-1, 0, 1],
        [-2, 0, 2],
        [-1, 0, 1]
    ], dtype=np.float64)
}

kernel_y = {
    'Robert': np.array([
        [0, 0, 0],
        [0, 0, 1],
        [0, -1, 0]
    ], dtype=np.float64),
    'Prewitt': np.array([
        [-1, -1, -1],
        [0, 0, 0],
        [1, 1, 1]
```

```
    ], dtype=np.float64),
    'Sobel': np.array([
        [-1, -2, -1],
        [0, 0, 0],
        [1, 2, 1]
    ], dtype=np.float64)
}
```

Robert filter

[9]:
```
for path in list_path_image :
    img = cv2.imread(path)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    grad_x = cv2.filter2D(img, cv2.CV_64F, kernel_x['Robert'])
    grad_y = cv2.filter2D(img, cv2.CV_64F, kernel_y['Robert'])
    img_gradient = np.abs(grad_x) + np.abs(grad_y)
    threshold = np.max(img)*0.33
    img_gradient[img_gradient < threshold] = 0
    img_gradient[img_gradient != 0] = 255
    img_gradient = cv2.normalize(img_gradient, None, 0, 255, cv2.NORM_MINMAX)
    img_gradient = np.uint8(img_gradient)

    img_array = [img, img_gradient, img + img_gradient]

    plt.figure(figsize=(15, 5))

    for i in range(3):
        plt.subplot(1, 3, i+1)
        plt.imshow(img_array[i], cmap='gray')
        plt.axis('off')
        plt.title(['Original', 'Gradient', 'Sharpen'][i])
    plt.show()
```



Original

Gradient

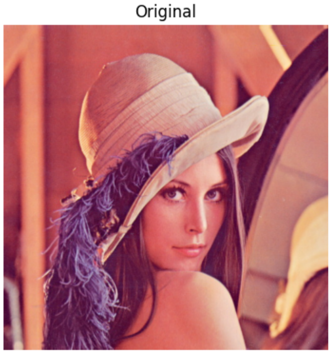
Sharpen

Original

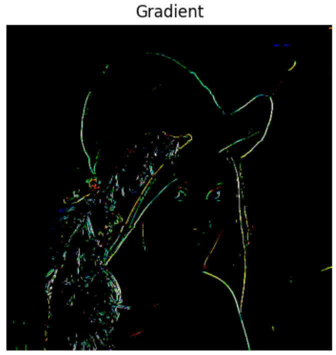
Gradient

Sharpen

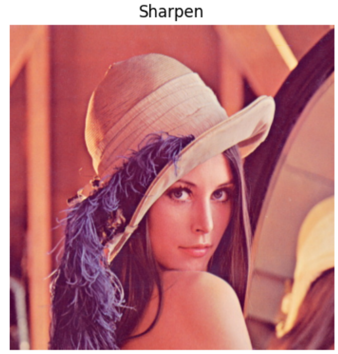
Original



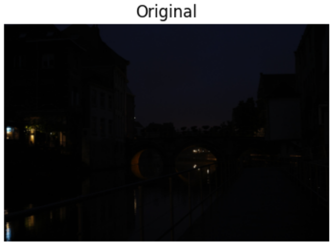
Gradient



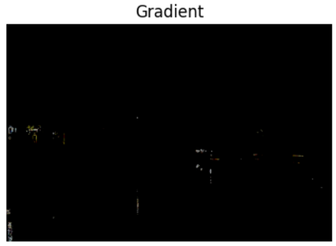
Sharpen



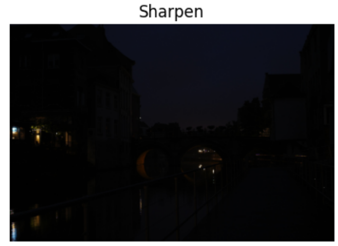
Original



Gradient

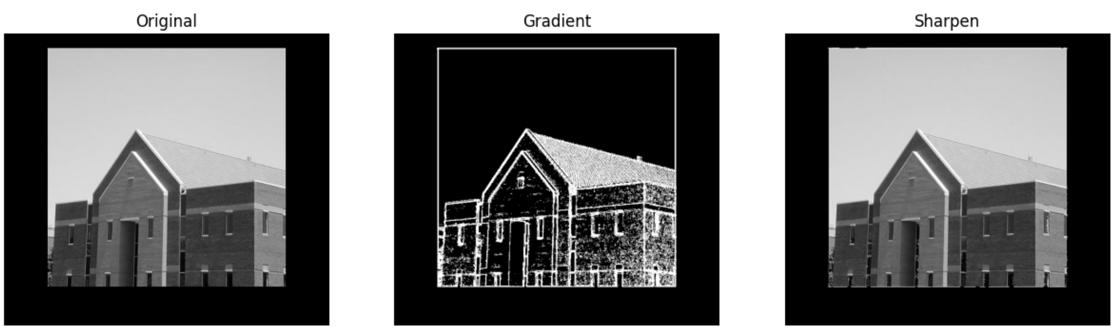


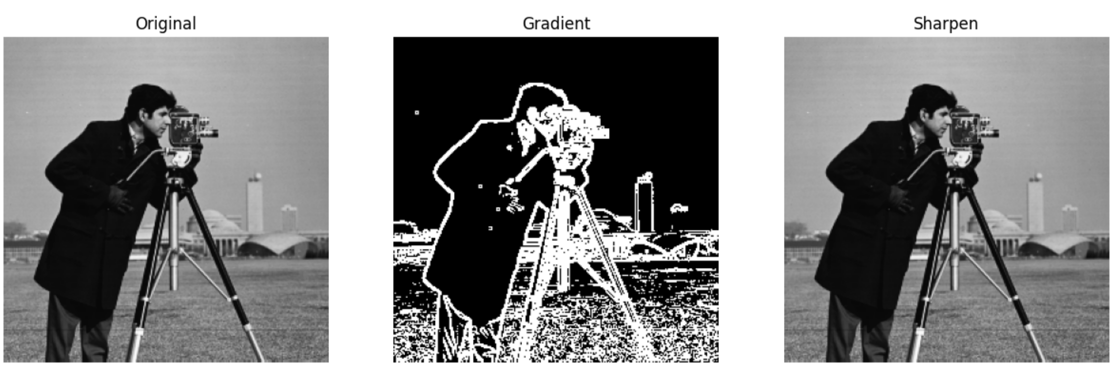
Sharpen

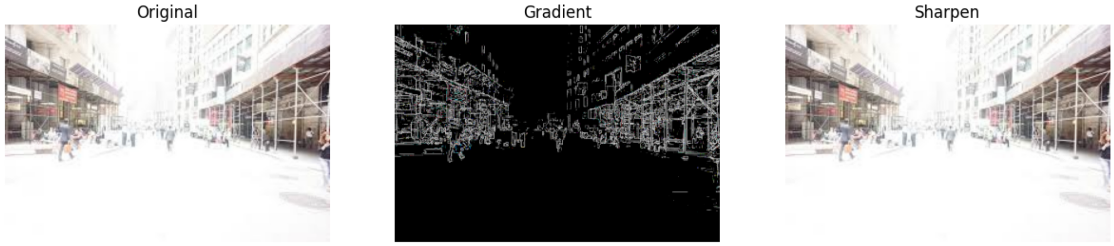


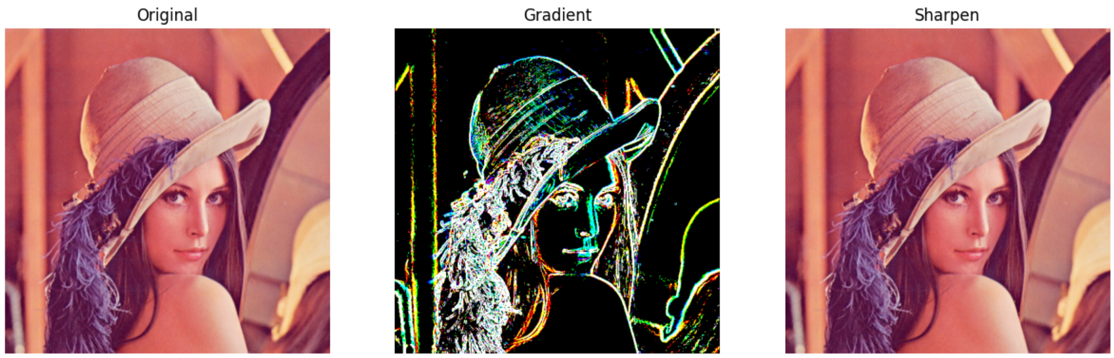
Sobel Filter

[10]:
```python
for path in list_path_image :
    img = cv2.imread(path)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    grad_x = cv2.filter2D(img, cv2.CV_64F, kernel_x['Sobel'])
    grad_y = cv2.filter2D(img, cv2.CV_64F, kernel_y['Sobel'])
    img_gradient = np.abs(grad_x) + np.abs(grad_y)
    threshold = np.max(img)*0.33
    img_gradient[img_gradient < threshold] = 0
    img_gradient[img_gradient != 0] = 255
    img_gradient = cv2.normalize(img_gradient, None, 0, 255, cv2.NORM_MINMAX)
    img_gradient = np.uint8(img_gradient)

    img_array = [img, img_gradient, img + img_gradient]
    plt.figure(figsize=(15, 10))
    for i in range(3):
        plt.subplot(1, 3, i+1)
        plt.imshow(img_array[i], cmap='gray')
        plt.axis('off')
        plt.title(['Original', 'Gradient', 'Sharpen'][i])
    plt.show()
```





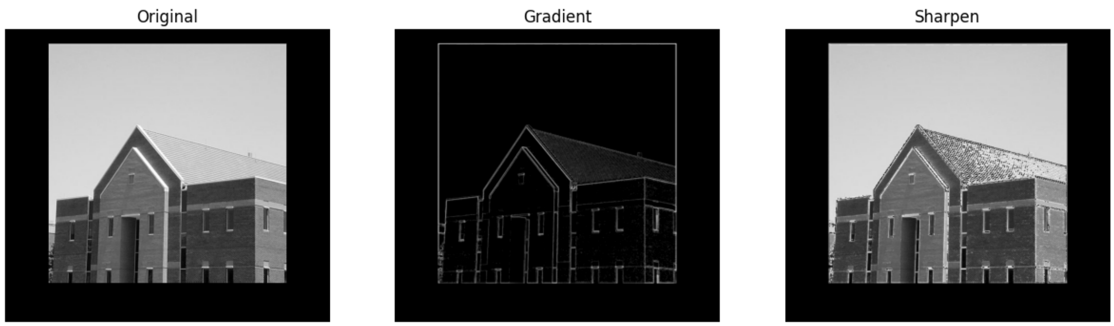


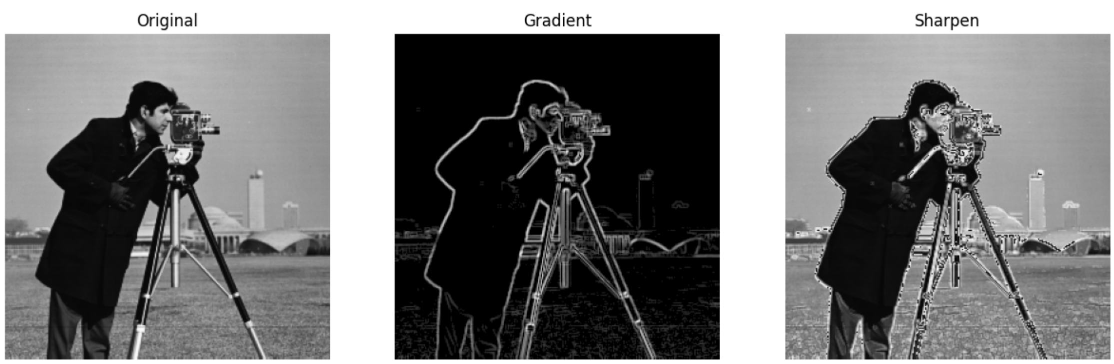


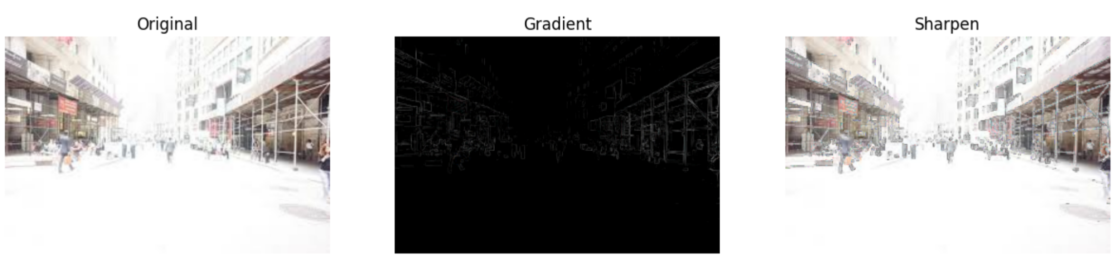
Prewitt Filter

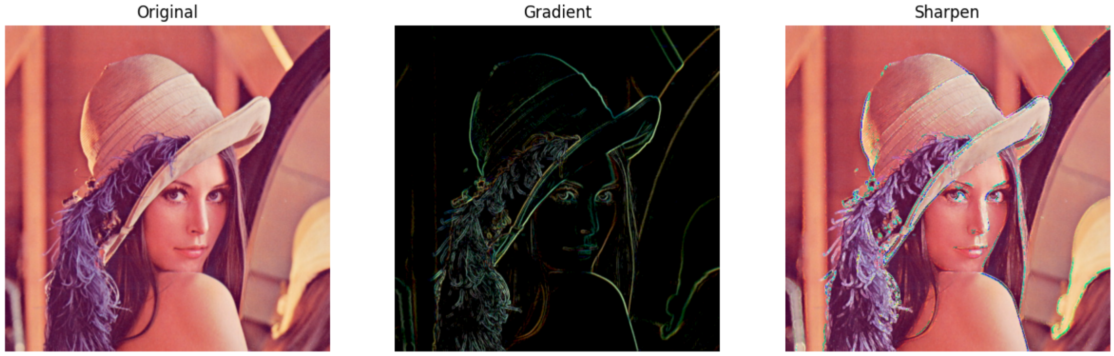
[12]:
```python
for path in list_path_image :
    img = cv2.imread(path)
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)
    grad_x = cv2.filter2D(img, cv2.CV_64F, kernel_x['Prewitt'])
    grad_y = cv2.filter2D(img, cv2.CV_64F, kernel_y['Prewitt'])
    img_gradient = np.abs(grad_x) + np.abs(grad_y)
    threshold = np.max(img)*0.33
    img_gradient[img_gradient < threshold] = 0
    img_gradient = cv2.normalize(img_gradient, None, 0, 255, cv2.NORM_MINMAX)
    img_gradient = np.uint8(img_gradient)

    img_array = [img, img_gradient, img + img_gradient]
```

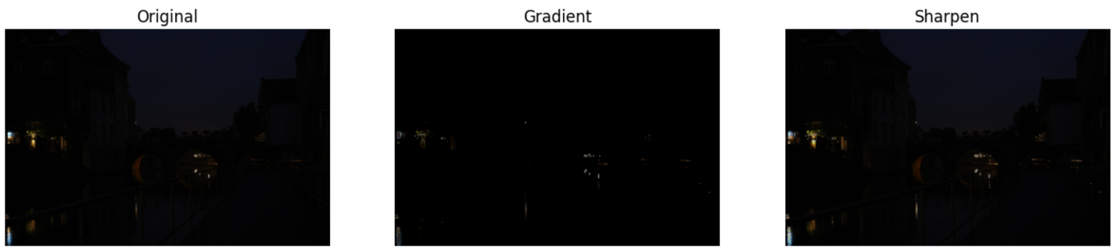
```python
plt.figure(figsize=(15, 10))
for i in range(3):
    plt.subplot(1, 3, i+1)
    plt.imshow(img_array[i], cmap='gray')
    plt.axis('off')
    plt.title(['Original', 'Gradient', 'Sharpen'][i])
plt.show()
```








Original
Gradient
Sharpen


Original
Gradient
Sharpen